
Final Project Report MATH 563

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Abstract

give some background

1 Introduction

1.1 Notation

Throughout this paper, we will denote the *proximal operator* of f with parameter λ by

$$\mathbf{prox}_{\lambda f}(v) := \arg \min_x \left\{ f(x) + \frac{1}{2\lambda} \|x - v\|_2^2 \right\} \quad (1)$$

The *indicator function* for a convex set C will be defined as

$$\delta_C(x) := \begin{cases} 0, & \text{if } x \in C \\ +\infty, & \text{if } x \notin C \end{cases} \quad (2)$$

We also denote the *Euclidean Projection* on a set C by

$$\mathbf{P}_C(v) := \arg \min_{x \in C} \|x - v\|_2 \quad (3)$$

1.2 Preliminaries

We first note an important relation between the proximal and subdifferential operators.

Proposition 1. Given a function $f \in \Gamma$, we have

$$\mathbf{prox}_{\lambda f} = (I + \lambda \partial f)^{-1}, \quad (4)$$

where $(I + \lambda \partial f)^{-1}$, for $\lambda > 0$, is called the **resolvent** of the operator ∂f . In particular, the proximal operator is the resolvent of the subdifferential operator [3].

Definition 1 (Maximal Monotone operator). A monotone operator is **maximal** if its graph is not properly contained in the graph of any other monotone operator. If A is monotone (resp. maximal monotone) then so are A^{-1} and λA if $\lambda > 0$.

The next proposition, given in [2], will be useful in the construction of the optimizing algorithms.

Proposition 2. If F is maximal monotone, then $(I + \lambda F)^{-1}(\hat{x})$ exists and is unique for all $\hat{x} \in R^n$. The value $x = (I + \lambda F)^{-1}(\hat{x})$ of the resolvent is the unique solution of the monotone inclusion

$$\hat{x} \in x + \lambda F(x) \quad (5)$$

Proposition 3 (Proximal operator splitting). If f is separable across two variables, so $f(x, y) = \varphi(x) + \psi(y)$, then

$$\mathbf{prox}_f(x, y) = (\mathbf{prox}_\varphi(x), \mathbf{prox}_\psi(y)) \quad (6)$$

Thus, evaluating the proximal operator of a separable function reduces to evaluating the proximal operators for each of the separable parts, which can be done independently [3].

Proposition 4. (Proximal operator of norms) If $f = \|\cdot\|$ is a general norm, then f^* is the indicator function on the unit ball for the dual norm $\|\cdot\|_*$, namely

$$\|\cdot\|^* = \delta_{\|\cdot\|_* \leq 1}$$

where $\|\cdot\|_*$ is given by

$$\|v\|_* = \sup_x \{\langle v, x \rangle \mid \|x\| \leq 1\}$$

In particular, the *Moreau Decomposition Theorem* gives that

$$\mathbf{prox}_{\|\cdot\|}(v) = \mathbf{P}_{\{x: \|x\|_* \leq 1\}}(v) - v$$

2 Algorithms

The algorithms we present are used to solve the non-blind image deblurring problem, which is a convex optimization problem of the form

$$\begin{aligned} \min_{x, y} & f(x) + g(y) \\ \text{subject to} & Ax = y, \end{aligned} \quad (7)$$

where $f, g \in \Gamma$ have inexpensive prox-operators. We are given a vectorized, blurred image b , along with the kernel K used to blur it using the model

$$Kx + \eta = b, \quad (8)$$

where x is the original image, and η the added noise. Our goal is to recover the original image x . We formulate the problem in two ways, namely, one which uses the L^1 norm, and the other the L^2 norm. First,

$$\min_x \|Kx - b\|_1 + \delta_S(x) + \gamma \|Dx\|_{\text{iso}}, \quad (9)$$

and second,

$$\min_x \|Kx - b\|_2^2 + \delta_S(x) + \gamma \|Dx\|_{\text{iso}}, \quad (10)$$

where $S = \{x \mid 0 \leq x \leq 1\}$, so that δ_S aims to restrict the range of allowed pixels. The iso norm is defined as

$$\|(x, y)\|_{\text{iso}} = \sum_{k=1}^n (x_k^2 + y_k^2)^{\frac{1}{2}}$$

2.1 Primal Douglas-Rachford splitting (Spingarn's method)

This first algorithm, which we will refer to by PDRS, was invented in the 1950s as a method to solve the heat equation [1]. Per (9), let

$$\begin{aligned} f(x) &= \delta_S(x) \\ g(x) &= \|Kx - b\|_1 + \gamma \|Dx\|_{\text{iso}} \end{aligned} \quad (11)$$

and consider the minimization problem

$$\min_{x, y} f(x) + g(y) + \delta_0 \left(\begin{bmatrix} A & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \right) \quad (12)$$

with the optimality condition

$$0 \in \partial f(x) + \partial g(y) + \partial \delta_0 \left(\begin{bmatrix} A & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \right) =: \mathcal{F}(x, y) \quad (13)$$

As stated in the literature [2], $\mathcal{F}$ is the subdifferential of closed, convex functions, hence is a maximal monotone operator. Thus, by (5), the zeros of $\mathcal{F}$ are the fixed points of the resolvent of $\mathcal{F}$. Therefore, if we can find the fixed points of $\mathcal{F}$, we'll have found the optimal points. Unfortunately, this particular problem is too computationally expensive, so the strategy is to split $\mathcal{F}$ into two maximal monotone operators $\mathcal{A}$ and $\mathcal{B}$ which have inexpensive proximal operators. Finding the fixed points of $\mathcal{A} + \mathcal{B}$ will be equivalent to finding that of $\mathcal{F}$. Define $\mathcal{A}$ and $\mathcal{B}$ as,

$$\begin{aligned}\mathcal{A}(x, y) &= \partial f(x) + \partial g(y) = \partial(\delta_S(x)) + \partial(\|Ky - b\|_1 + \gamma\|Dy\|_{\text{iso}}) \\ \mathcal{B}(x, y) &= \partial \delta_0 \left(\begin{bmatrix} K \\ D \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \right)\end{aligned}\quad (14)$$

We now focus on finding the resolvents of $\mathcal{A}$ and $\mathcal{B}$,

$$\begin{aligned}J_{\mathcal{A}} &= (I + \lambda \mathcal{A})^{-1} = \text{prox}_{\lambda(f+g)}(x, y) \quad [\text{by (4)}] \\ &= (\text{prox}_{\lambda f}(x), \text{prox}_{\lambda g}(y)) \quad [\text{by Prop. 3}]\end{aligned}\quad (15)$$

where

$$\text{prox}_{\lambda f}(x) = \text{prox}_{\lambda \delta_S}(x) = \arg \min_u \frac{1}{2} \|x - u\|_2^2 + \delta_S(x) = \arg \min_{u \in S} \frac{1}{2} \|x - u\|_2^2 = \mathbf{P}_S(x), \quad (16)$$

which is the euclidean projection operator onto S . It can be computed element-wise by

$$(\mathbf{P}_S(x))_i = \begin{cases} 0, & \text{if } x_i < 0 \\ x_i, & \text{if } x_i \in S = \max\{0, \min\{x_i, 1\}\} \\ 1, & \text{if } x_i > 1 \end{cases} \quad (17)$$

Now, we observe that $g(y)$ described in (11) can be written in a different form by letting

$$g(y_1, y_2, y_3) = \|y_1 - b\|_1 + \gamma\|(y_2, y_3)\|_{\text{iso}} \quad (18)$$

subject to

$$y_1 = Ky \quad \text{and} \quad \begin{bmatrix} y_2 \\ y_3 \end{bmatrix} = Dy \iff \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} K \\ D \end{bmatrix} y$$

Then, by Proposition 3, notice that the proximal operator of g may be obtained by splitting its components into

$$\varphi(y_1) = \|y_1 - b\|_1 \quad \text{and} \quad \psi(y_2, y_3) = \gamma\|(y_2, y_3)\|_{\text{iso}}$$

which gives

$$\text{prox}_{\lambda g}(y_1, y_2, y_3) = (\text{prox}_{\lambda \varphi}(y_1), \text{prox}_{\lambda \psi}(y_2, y_3)) \quad (19)$$

We know that the proximal operator of the L^1 norm is the soft-thresholding operator T_λ , so $\text{prox}_{\lambda \varphi}(y_1)$ is given element-wise as follows,

$$(\text{prox}_{\lambda \varphi}(y_1))_i = (\text{prox}_{\lambda \|\cdot\|_1}(y_1 - b))_i = T_\lambda((y_1 - b)_i) := \begin{cases} (y_1 - b)_i - \lambda, & \text{if } (y_1 - b)_i > \lambda \\ (y_1 - b)_i + \lambda, & \text{if } (y_1 - b)_i < -\lambda \\ 0, & \text{otherwise.} \end{cases} \quad (20)$$

As a result of Proposition 4, $\text{prox}_{\lambda \psi}(y_2, y_3)$, is given element-wise by

$$(\text{prox}_{\lambda \psi}(y_2, y_3))_i = \alpha_i((y_2)_i, (y_3)_i),$$

where,

$$\alpha_i = \begin{cases} 1 - \frac{\lambda}{\sqrt{(y_2)_i^2 + (y_3)_i^2}}, & \text{if } \sqrt{(y_2)_i^2 + (y_3)_i^2} > \lambda \\ 0, & \text{otherwise.} \end{cases}$$

(provide a proof of this)

As for the resolvent of $\mathcal{B}$, it corresponds to the projection (x, y) of a vector $(\bar{x}, \bar{y})$ onto the subspace $\{(x, y) \mid \begin{bmatrix} K & D \end{bmatrix}^T x = y\}$, expressed as

$$J_{\mathcal{B}} = (I + \lambda \mathcal{B})^{-1} = \left(x = \left(I + \begin{bmatrix} K & D \end{bmatrix} \begin{bmatrix} K \\ D \end{bmatrix} \right)^{-1} \left(\bar{x} + \begin{bmatrix} K \\ D \end{bmatrix} \bar{y} \right), y = \begin{bmatrix} K \\ D \end{bmatrix} x \right)$$

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Input           : Blurred image  $b$ , step size  $t$ , relaxation parameter  $\rho$ 
Output          : Deblurred image  $x$ 
Initialize      :  $(z_1^0, z_2^0)$ 
for  $k = 1, 2, \dots$ , do
     $x^k \leftarrow \text{prox}_{\mathcal{I}_f}(z_1^{k-1})$ ;           // Performs resolvent of  $\mathcal{A}$ 
     $y^k \leftarrow \text{prox}_{\mathcal{I}_g}(z_2^{k-1})$ ;
     $u^k \leftarrow (I + A^T A)^{-1} (2x^k - z_1^{k-1} + A^T (2y^k - z_2^{k-1}))$ ; // Performs resolvent of  $\mathcal{B}$ 
     $v^k \leftarrow A(u^k)$ ;
     $z_1^k \leftarrow z_1^{k-1} + \rho(u^k - x^k)$ ;
     $z_2^k \leftarrow z_2^{k-1} + \rho(v^k - y^k)$ ;
end
return  $x\text{Sol} = \text{prox}_{\mathcal{I}_f}(z^k)$ ;

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In this algorithm, we consider the dual problem,

$$\text{maximize } -f^*(-A^T z) - g^*(z) \quad (21)$$

with the optimality conditions for both the primal and the dual problem

$$0 \in \begin{bmatrix} 0 & 0 & A^T & I \\ 0 & 0 & -I & 0 \\ -A & I & 0 & 0 \\ -I & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} + \begin{bmatrix} 0 \\ \partial g(y) \\ 0 \\ \partial f^*(w) \end{bmatrix}$$

In the above, w is the dual variable and z is the dual multiplier for the constraint $y = Ax$, thus the dual problem is subject to

$$A^T z + w = 0$$

We reduce this to a smaller system by using the relationship $\partial g^* = (\partial g)^{-1}$, which follows from the fact that g is closed and convex, and by the properties of the subgradient and conjugate. We can eliminate y by using the above equality that $\partial g^* = (\partial g)^{-1}$, which would leave us with

$$0 \in \begin{bmatrix} 0 & A^T & I \\ -A & 0 & 0 \\ -I & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ z \\ w \end{bmatrix} + \begin{bmatrix} 0 \\ \partial g^*(z) \\ \partial f^*(w) \end{bmatrix}$$

We can further reduced this by eliminating w , using a similar equality that $\partial f = (\partial f^*)^{-1}$, we get

$$0 \in \begin{bmatrix} 0 & A^T \\ -A & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} \partial f(x) \\ \partial g^*(z) \end{bmatrix} \quad (22)$$

We can now split the monotone operator we have above to a sum of two monotone operators given as

$$\mathcal{A}(x, z) = \begin{bmatrix} \partial f(x) \\ \partial g^*(x) \end{bmatrix} \quad \mathcal{B}(x, z) = \begin{bmatrix} 0 & A^T \\ -A & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} \quad (23)$$

Similar to the previous algorithm, we now find the resolvents of $\mathcal{A}$ and $\mathcal{B}$.

As seen before with the resolvent of $\mathcal{A}$, the value $(x, y) = (I + At)^{-1}(\hat{x}, \hat{z})$ is given by

$$x = \mathbf{prox}_{\lambda f}(\hat{x}) \quad z = \mathbf{prox}_{g^*}(\hat{z})$$

For z , we can apply moreau decomposition to get an alternate expression dealing with the conjugate g^* , thus we have,

$$z = \hat{z} - t \mathbf{prox}_{t^{-1}g} \left(\frac{\hat{z}}{t} \right)$$

Simplifying the above, we are left with,

$$z = \hat{z} - \mathbf{prox}_{gt}(\hat{z})$$

This will be a similar computation to the prox operator in algorithm 1, but we have right scalar multiplication in this case.

The resolvent for $\mathcal{B}$ is a linear mapping, so considering resolvents for skew-symmetric linear operators

$$\begin{aligned} (I + t\mathcal{B})^{-1} &= \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} I \\ tA \end{bmatrix} (I + t^2 A^T A)^{-1} \begin{bmatrix} I \\ -tA \end{bmatrix}^T \\ &= \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -tA^T \\ I \end{bmatrix} (I + t^2 A A^T)^{-1} \begin{bmatrix} tA^T \\ I \end{bmatrix}^T \end{aligned}$$

Letting (w, v) be the value of the resolvent of $\mathcal{B}$,

$$\begin{aligned} \begin{bmatrix} w \\ v \end{bmatrix} &= \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} I \\ tA \end{bmatrix} (I + t^2 A^T A)^{-1} \begin{bmatrix} I \\ -tA \end{bmatrix}^T \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \end{aligned}$$

Here is the Primal-Dual Douglas Rachford splitting algorithm,

Algorithm 2: Primal-Dual Douglas-Rachford Splitting

Input : Blurred image b , step size t , relaxation parameter ρ

Output : Deblurred image x

Initialize : p^0, q^0

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1 for  $k = 1, 2, \dots$ , do
2    $x^k \leftarrow \mathbf{prox}_{tf}(p_1^{k-1})$ ; // Performs resolvent of  $\mathcal{A}$ 
3    $z^k \leftarrow \mathbf{prox}_{tg^*}(q^{k-1})$ ;
4    $\begin{bmatrix} w^k \\ v^k \end{bmatrix} = \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} \begin{bmatrix} 2x^k - p^{k-1} \\ 2z^k - p^{k-1} \end{bmatrix}$ ; // Performs resolvent of  $\mathcal{B}$ 
5    $p^k \leftarrow p^{k-1} + \rho(w^k - x^k)$ ;
6    $q^k \leftarrow q^{k-1} + \rho(v^k - z^k)$ ;
7 end
8 return  $xSol = \mathbf{prox}_{tf}(p^k)$ ;

```

Acknowledgments

Please include a list of each person (with student ID) in your group and write a paragraph **for each person** with exactly what they did. Note that the Acknowledgment section does not count towards the page limit.

Example Acknowledgments would look like:

Alexandre St-Aubin (Student ID: 261115785) We ball.

Jonathan Campana (Student ID: 260985613) We ball.

Elliot Forcier-Poirier (Student ID:) We ball.

Edmund Wu (Student ID:) We ball.

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A Appendix

Optionally include extra information (complete proofs, additional experiments and plots) in the appendix. This section will often be part of the supplemental material.